

BEYOND THE HORIZON

SKYGAZE INDIA



Also Inside: Astronomical Events | Astrophotographs
Lunar Calendar | Galaxies | Dark Matter | Multiverse Theory

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INTRODUCTION

Beyond the Horizon, Skygaze India's monthly publication, is meticulously curated to stimulate intellectual curiosity and foster deeper public engagement with astronomy and space science.

Each issue presents an amalgamation of celestial event highlights, exemplary astrophotography features, perspicacious articles, and exclusive content contributed by our community of astronomy enthusiasts. Our objective is to make space science both accessible and profoundly inspiring.

OUR TEAM

Founder & Editor in Chief

Harsh

Contributors

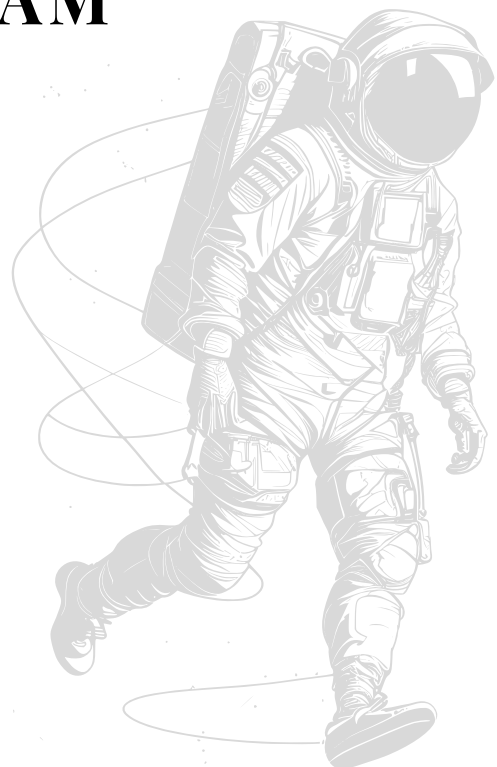
Mina Gerges, UAE

Jheel Jain, India

Sanjana Milind, India

Design & Layout

Kanak Garg



"At Skygaze India, our mission is to make space science accessible—to bring the cosmos closer by turning complex knowledge into everyday curiosity for all."

Harsh
Founder & CEO

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ASTRONOMICAL EVENTS

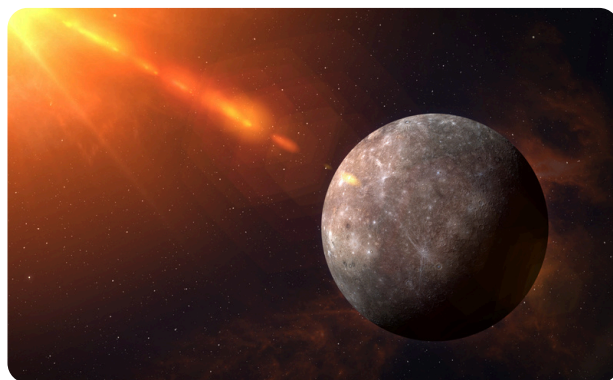


DECEMBER 5: COLD FULL MOON

The Cold Moon will brighten the December skies, marking the final full moon of the year. Its crisp, silvery light casts long winter shadows—perfect for moonlit photography or a peaceful evening walk beneath the glowing orb. This luminous full moon has long been associated with the deepening chill of early winter.

DECEMBER 8: MERCURY AT GREATEST ELONG

Mercury reaches its greatest eastern elongation, offering one of the best chances this year to spot the elusive planet. Low in the western sky just after sunset, Mercury will appear as a sharp, bright point. With clear horizons, observers can enjoy rare visibility of the innermost planet before it dips quickly toward the horizon.



DECEMBER 13-14: GEMINIDS METEOR SHOWER

One of the strongest and most colorful meteor showers of the year, the Geminids will peak on these nights. With up to 120 meteors per hour under optimal dark skies, the Geminids produce bright, bold streaks that often show subtle hues. The shower radiates from the constellation Gemini, offering a spectacular winter sky display.



DECEMBER 21 - WINTER SOLSTICE

The longest night of the year arrives as the Sun reaches its southernmost point in the sky. The winter solstice marks the official start of winter in the Northern Hemisphere. Stargazers can take advantage of the extended darkness for extended observation sessions, while many celebrate the slow return of longer days ahead.





DECEMBER 22-23 - URSIDS METEOR SHOWER

The Ursids, a modest yet charming meteor shower, will peak on these nights. Typically producing 5–10 meteors per hour, the shower radiates from the constellation Ursa Minor. Though not as prolific as the Geminids, the Ursids occasionally surprise with brief bursts of activity, best viewed under clear, dark winter skies. A subtle but rewarding shower for those braving the crisp winter night.

LUNAR CALENDAR

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	01  85.37% Waxing Gibbous 11.08 day	02  92.79% Waxing Gibbous 12.21 day	03  97.8% Full Moon 13.37 day	04  99.94% Full Moon 14.53 day	05  99.02% Full Moon 15.7 day	06  95.2% Waning Gibbous 16.84 day
07  88.9% Waning Gibbous 17.96 day	08  80.75% Waning Gibbous 19.04 day	09  71.37% Waning Gibbous 20.07 day	10  61.35% Last Quarter 21.07 day	11  51.18% Last Quarter 22.04 day	12  41.26% Last Quarter 22.97 day	13  31.9% Last Quarter 23.89 day
14  23.35% Waning Crescent 24.79 day	15  15.85% Waning Crescent 25.68 day	16  9.6% Waning Crescent 26.57 day	17  4.78% Waning Crescent 27.46 day	18  1.57% New Moon 28.35 day	19  0.09% New Moon 29.25 day	20  0.44% New Moon 0.62 day
21  2.66% New Moon 1.54 day	22  6.73% Waxing Crescent 2.47 day	23  12.6% Waxing Crescent 3.41 day	24  20.1% Waxing Crescent 4.37 day	25  29.03% Waxing Crescent 5.35 day	26  39.08% First Quarter 6.35 day	27  49.88% First Quarter 7.37 day
28  60.96% First Quarter 8.42 day	29  71.77% Waxing Gibbous 9.5 day	30  81.65% Waxing Gibbous 10.6 day	31  89.94% Waxing Gibbous 11.73 day			

NOVEMBER RECAP

STARGAZING AT GD GOENKA GLOBAL SCHOOL

On 4th November 2025, Skygaze India conducted an engaging astronomy outreach session at **GD Goenka Global School, Gurugram**, turning the campus courtyard into a vibrant open-air learning space for young explorers.

With the help of professional-grade telescopes, students and faculty members observed the Moon, Saturn, and bright stars scattered across the evening sky — many witnessing Saturn's iconic rings for the first time.

The enthusiasm and curiosity shown by the participants made the event truly memorable, leaving the school community excited for upcoming Skygaze India astronomy sessions.



"We had a wonderful experience with Skygaze India. The children and our staff reached peak excitement after seeing the views of the Moon and Saturn's rings. It was not only a great learning experience but also a fun-filled event. We would love to organise more such sessions in the future."

~Ms. Navina Chadha,
Principal



STARGAZING AT GENESIS GLOBAL SCHOOL

On 7th November 2025, Skygaze India hosted an engaging night sky observation session at **Genesis Global School, Noida**, where 20 enthusiastic students and over 5 faculty members gathered for an evening of celestial exploration.

Guided by Skygaze facilitators, participants observed the Moon and Saturn through a high-powered telescope, witnessing the planet's iconic rings and the Moon's detailed craters with remarkable clarity — a first-time experience for many.

The excitement and curiosity shared by the students made the session truly memorable, reinforcing Skygaze India's commitment to bringing hands-on astronomy learning to classrooms across the country.

"It was a heart-touching experience to see our children experience the Moon and Saturn. Their curiosity and enthusiasm made the session truly special."

~Faculty from Genesis Global School



SKYGAZE INDIA BECOMES AN ISRO-REGISTERED SPACE TUTOR

Skygaze India has achieved a significant milestone by being officially recognized as an **ISRO-Registered Space Tutor** under the Capacity Building Programme Office (CBPO), ISRO Headquarters.

This prestigious recognition empowers Skygaze India to **conduct certified space education programs, workshops, observation sessions, and national-level outreach activities** across India.

Through years of astronomy outreach—telescope observation events, school programs, stargazing experiences, and the Skygaze Academy learning platform—Skygaze India has continuously inspired curiosity and scientific thinking among students, educators, and the public.

With ISRO's endorsement, the organization will expand its offerings to include:

- ISRO-aligned space education workshops
- Hands-on astronomy and astrophysics programs for schools
- Certified telescope observation camps
- Public engagement events promoting scientific curiosity
- Training modules for teachers and educators
- Learning programs aligned with ISRO's educational framework

"Becoming an ISRO-recognized Space Tutor is a moment of immense pride for us. This recognition strengthens our mission of democratizing space education and bringing astronomy closer to every learner. We look forward to expanding our outreach and building meaningful collaborations with institutions across India."

~Harsh Verma, Founder, Skygaze India

What This Recognition Means

Skygaze India now joins a select group of organizations trusted by ISRO to deliver **authentic, high-quality space education nationwide**.

This achievement opens pathways for broader school collaborations, certified workshops, STEM awareness programs, and impactful public engagement in astronomy and space science.



ASTROPHOTOGRAPHS



The Beaver Full Moon captured by Mr. Harsh, Founder at Skygaze India, Delhi, India

FEATURE YOUR WORK NEXT MONTH !

Have an astrophotograph you'd love to share with the world?

We're inviting submissions for our next issue of Beyond the Horizon, a Skygaze India publication.

Submit via email on support@skygazeindia.com



IS OUR UNIVERSE REALLY OF THE COLOURS WE SEE IN IMAGES? OR IS IT A MAGNIFICENT LIE?

Surfing through social media every day, it's hard not to pause when you see breathtaking images of the universe — vibrant galaxies, radiant stars, and nebulae that look like cosmic artwork. These dazzling visuals mesmerize us, painting an incredible picture of the cosmos in our minds. But amidst this awe, one question often lingers: Are these the real colours of galaxies, or just an elaborate artistic fabrication?

The answer lies somewhere between “yes” and “no.” It's a fascinating mix of scientific reality and creative interpretation. Let's dive deeper into this cosmic mystery.

The Science Behind the colours

Our vision is not complete

The human eye can only perceive a small part of the electromagnetic spectrum — the visible spectrum, which ranges from 380 to 700 nanometers. This is why we see colours from violet to red. However, the universe emits light across a vast range of wavelengths, most of which are invisible to us.

Infrared light reveals cooler objects, such as star-forming regions and dust clouds.

Ultraviolet light showcases energetic phenomena like young stars and starburst galaxies.

X-rays and gamma rays expose high-energy events like black hole activity or supernovae.

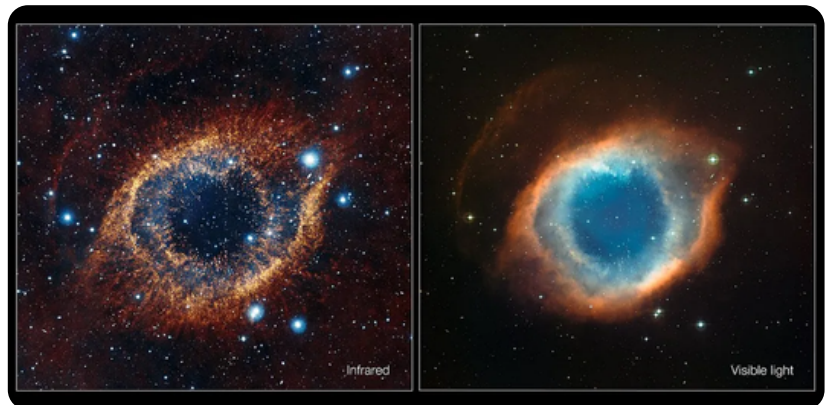


Image of Helix Nebula in Infrared and Visible spectrum



Pillars of creation through Hubble telescope & JWST

Radio waves unveil the cold hydrogen gas crucial for galaxy formation.

Since most cosmic light is outside the visible spectrum, astronomers use specialized telescopes to detect it and translate it into visible colours, revealing the universe in ways our eyes could never see.

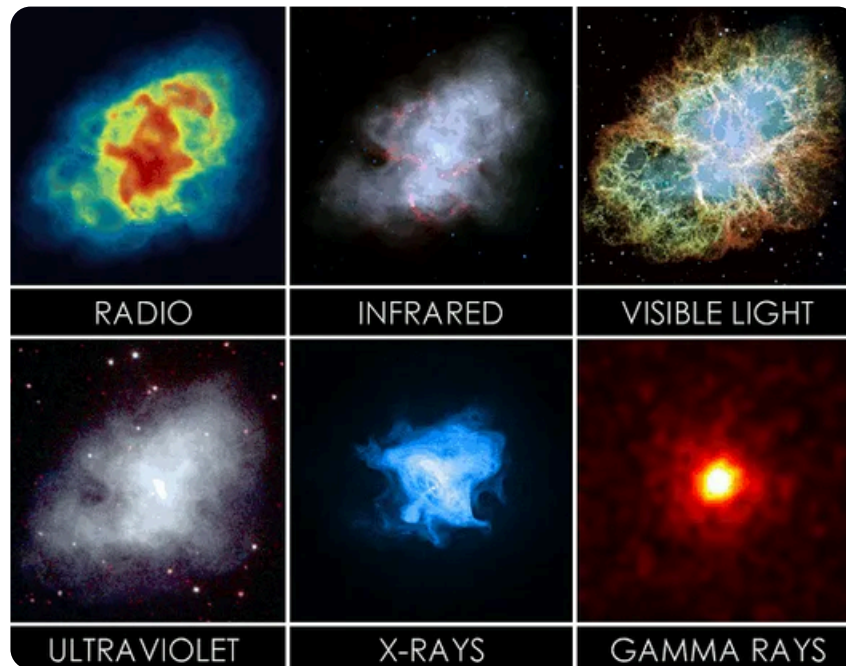


Image of Crab nebula in different spectrums

False Colour Imaging: A Tool to See the Unseen

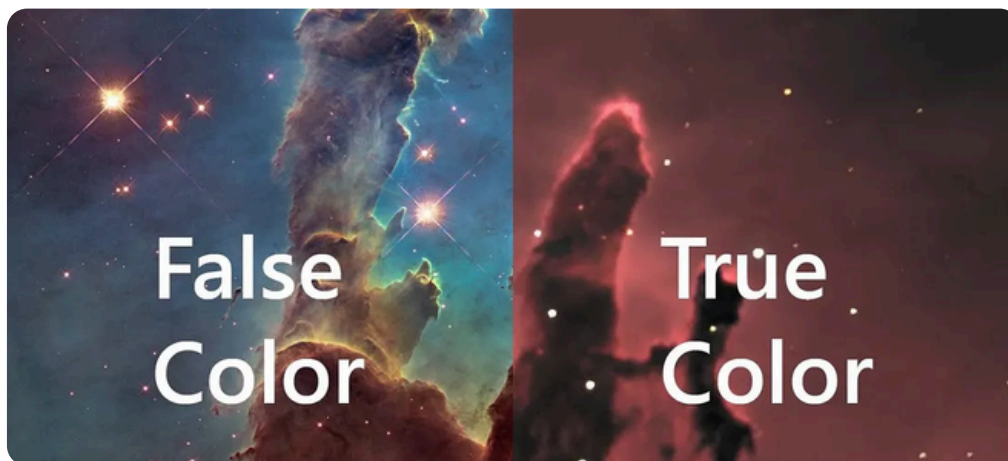
To visualize invisible wavelengths, astronomers use false colour imaging, where specific colours are assigned to wavelengths beyond our vision. This technique isn't about deception; it's about revealing the unseen.

For example:

Infrared light might be depicted as red to highlight warm dust and star-forming regions. Ultraviolet light might appear as blue or violet to represent energetic young stars. X-rays and gamma rays might be displayed in vivid hues like purple or green to emphasize extreme cosmic events.

This method helps astronomers' study complex structures and phenomena that would otherwise remain hidden. By mapping these invisible signals, false colour imaging transforms data into a visually understandable format, bridging the gap between raw science and human perception.

The human eye can only capture light in real-time and struggles with faint objects, especially those billions of light-years away. Telescopes, however, can collect light over extended periods — hours, days, or even weeks. This process, called **long-exposure photography**, reveals the faintest details of distant galaxies, nebulae, and stars.



False colouring of Eagle Nebula

Space photos' colors: not just pretty, but clues to the universe!

Blue and white show us young, hot stars, often busy forming in spiral arms of galaxies. Red and yellow colours tell us about older, cooler stars, usually found in the central parts of galaxies or in elliptical galaxies.

And those cool, unique colours like green? Those are often glowing gas clouds, especially hydrogen gas that's been energized. They're great markers for where new stars are born, or even to spot specific elements like oxygen. So, each colour is like a piece of a puzzle, helping scientists figure out how stars live and die, what galaxies are made of, and how the whole universe works!

Is It a Lie?

Far from being a "lie," the colors we see in galaxy images are representations of reality, crafted to convey the full story of the cosmos. While they may not be what our eyes would perceive directly, they're based on real data collected from telescopes. These colors are tools, helping us decode the universe's mysteries and visualize its hidden wonders.

Conclusion: A Magnificent Truth

The vibrant colors of galaxies are not just aesthetic choices; they're gateways to understanding the universe. Every hue reveals secrets about star formation, chemical processes, and cosmic evolution.

~Jheel Jain

Aspiring Astrophysicist (Rajasthan, India)

CONTRIBUTE TO BEYOND THE HORIZON!

Have an astronomy article, story, or thought to share?
We're inviting written submissions for our next publication!

Submit via email on support@skygazeindia.com



WHAT IS DARK MATTER REALLY MADE OF?

The particle identity of dark matter; hereafter DM is still unknown; however, it makes up about 85% of all our universe (Ellis et al. 1984). In this article we will discuss briefly three main ideas Weakly Interacting Massive Particles (WIMPs), axions, and sterile neutrinos. We cover the theoretical motivation, astrophysical role, strategies of detection and the most recent constraints in experimentation.

Introduction

DM was noticed firstly by astronomers because of the fast rotation of galaxies comparable to visible matter alone. More evidence also come from gravitational microlensing and cosmic microwave background (CMB). Scientists suggest that DM consists of new kind of particles, since it does not emit or absorb light. These particles extend beyond our known standard model (SM).

There are three candidates discussed here and are presented in Table 1.

1. Weakly Interacting Massive Particles (WIMPs)

1.1 Motivation & Production

It motivates connections beyond our known standard model. They are popular as their abundance were predicted for the big bang. This prediction matches the observed amount of DM(Collaboration et al. 2023).

1.2 Detection

They are detected via different methods such as

- (i) Direct Detection: nuclear recoil (elastic or inelastic scatters of galactic DM particles with atomic nuclei or electrons) in low background underground detectors,
- (ii) Indirect Detection: annihilation products form astrophysical targets
- (iii) Missing energy signatures from collider production (Bozorgnia et al. 2025).

WIMPs	Axions	Dark Photons	Sterile Neutrinos
Heavy particles that interact through the weak nuclear force.	Light particles with no electric charge and interact extremely weakly with ordinary matter.	Can be produced in the early universe through scattering or annihilation.	Versions of neutrinos that barely interact.

Table 1: Different types of DM candidates

1.3 Progress and constraints

Famous experiments such as LUX-ZEPLIN and XENONnT searched for WIMPs and atomic nuclei deep underground in tiny collisions (Aalbers et al. 2025). No signals have been detected due to low ranges of cross-sections. These results motivated the exploration of weaker coupling, sub-GeV masses as presented in Fig.1 and non-thermal production scenarios (Aalbers et al. 2023).

2. Axions

2.1 Motivation

Axion is a hypothetical elementary particle. This particle possesses a small mass with no electric charge. They were proposed mainly to solve the strong charge parity theoretical problem (CP)1 in Quantum Chromodynamics (QCD) (Choi and Righi 2024). Axions behave as invisible matter as they fit the basic required properties of DM: gravitating without emitting or absorbing light (Goodman et al. 2025).

2.2 Detection

Axions are coupled to photons weakly. Recent experiments such as ADMX2 converts axions into detectable photons via strong magnetic fields (Goodman et al. 2025).

2.3 Recent progress

ADMX excluded some possibilities of detecting axions in some mass and strengths of interaction ranges. Additionally, ADMX now is sensitive to examine some theoretical models such as DFSZ3 where those kind of axions are harder to be detected as they interact weakly.

3. Sterile Neutrinos

3.1 Motivation

Neutrinos are very light particles as singlet fermions (Kopp 2022). Sterile neutrinos are special class of them. These particles might decay producing a normal neutrinos and X-ray photons.

Bulbul et al.(2014) found an unexplained signal around 3.5 keV, which might be caused by sterile neutrinos as in figure 2, however, Dessert et al. (2020) ruled out this interpretation.

3.2 Detection & constraints

Sterile neutrinos can decay producing X-ray lines; however, simple production scenarios are constrained by X-ray searches and structure formation data.

Conclusion

No single candidate is confirmed, as all remain theoretically well motivated and under experimentation.

The nature if DM remains a puzzle and one of the most challenges in physics.

WIMP searches placed restrictions on interaction cross-section, also the absence of detection motivated wider exploration of lighter and weaker interacting particles.

Axions, offer testable predictions, meanwhile, sterile neutrinos offer a connection between dark matter and neutrino physics.

-Mina Gerges

Department of Physics,
College of Science,
United Arab Emirates University

THIS MONTH IN INDIAN SPACE HISTORY

DEC, 2019 11	PSLV-C48 / RISAT-2BR1 MISSION ISRO launched PSLV-C48 with the radar imaging satellite RISAT-2BR1 and several foreign satellites, boosting India’s all-weather Earth observation for surveillance and disaster management.
DEC, 2020 17	PSLV-C50 / CMS-01 COMMUNICATION SATELLITE The PSLV-C50 mission successfully placed the CMS-01 communication satellite into orbit, enhancing telecom and broadcasting services, especially in underserved regions, and strengthening India’s communication infrastructure.
DEC, 2025 05	PSLV-C59 / PROBA-3 MISSION (ESA COLLABORATION) ISRO’s PSLV-C59 launched ESA’s Proba-3 mission to study the Sun’s inner corona and improve space-weather understanding, underscoring India’s role as a reliable international launch partner.
DEC, 2025 07	PSLV-C60: INDIA’S FIRST SPACE DOCKING MISSION With the SpaDeX mission on PSLV-C60, India demonstrated key technologies for spacecraft rendezvous and docking, a vital capability for future space stations and servicing. This marks a major step toward India’s human spaceflight ambitions.

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